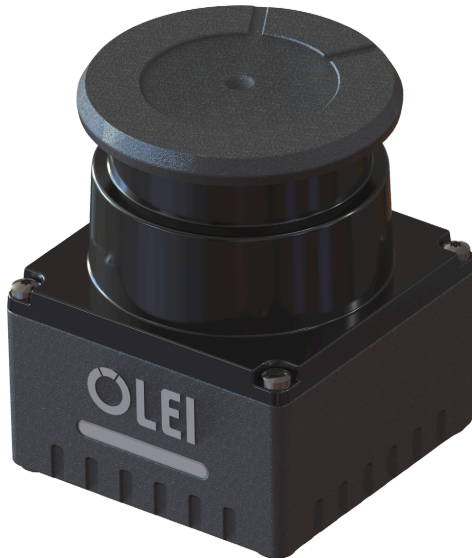


2D 360° SURVEY LIDAR · VFH SERIES

100K-POINT OLEI VFH1-50 360° LIDAR

Half a kilogram, under five watts, and 100 000 points a second across a full circle to 50 metres. The VFH1-50 is the LR-1F's reach in a smaller, lighter, denser-sampling package with NTP time sync and a UDP or TCP stream.



■ WHY THIS SENSOR

- **360° full-angle scanning to 50 m** — reflector and natural navigation from a single sensor
- **100 000 points per second at 25 Hz** — 60–100K depending on rate — the densest cloud in the 2D line
- **0.09° fine angle resolution** — held across 15, 20 and 25 Hz rather than traded against rate
- **< ±20 mm absolute accuracy from 2% to 80% reflectivity** — and on reflectors, with ≤ 20 mm repeat from 0.4 to 50 m
- **< 500 g in a 65 × 65 × 79.5 mm body** — roughly half the LR-1F's mass and volume
- **< 5 W at 12–24 V DC** — runs on a battery platform's budget with room to spare
- **NTP time synchronisation, UDP and TCP output** — point-cloud coordinates, reflectivity and intensity, timestamped
- **DC 5.5 × 2.1 power and RJ45 on 50 cm pigtails** — IP65, -10 to +50 °C, OleView point-cloud software

0.1–50 m

MEASURING RANGE

360°

FIELD OF VIEW

100K

POINTS / SECOND

< 500 g

WEIGHT

■ APPLICATION SCENARIOS



SURVEY & MAPPING

360° natural-feature and reflector mapping at range



HYBRID NAVIGATION

Reflector and natural navigation from one sensor



COMPACT PLATFORMS

Service robots and small AMRs with tight sensor pockets



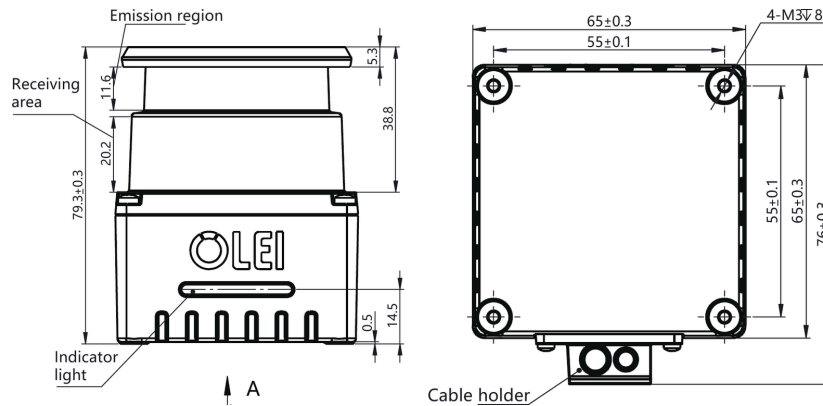
AREA MONITORING

Perimeter and equipment-envelope intrusion detection

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class I
Laser channels	1
Field of view (horizontal)	360°
Scanning frequency	15 / 20 / 25 Hz
Angle resolution	0.09°
Beam divergence	< 5 mrad
Horizontal error (horizontal angle)	≤ ±1°
Measuring range (based on reflectivity)	0.1–50 m @ 80% reflectivity 0.1–15 m @ 10% · 0.1–5 m @ 4% · 0.1–50 m @ reflector
Distance resolution	1 mm
Repeat accuracy	≤ 20 mm (0.4–50 m)
Absolute accuracy	< ±20 mm (2%–80% reflectivity) Reflector < ±20 mm
Point cloud density	100K @ 25 Hz

PARAMETER	TECHNICAL DATA
Ethernet output	100 Mbps Ethernet
Output data protocol	UDP, TCP
UDP packet content	Point cloud coordinates, reflectivity, intensity
Time synchronisation	NTP
Power supply	DC 5.5 × 2.1 power connector Pigtail, cable length 50 cm
Communication	RJ45 network port Pigtail, cable length 50 cm
Operating voltage	12–24 V
Power consumption	< 5 W
Mechanical dimensions	< 65 × 65 × 79.5 mm
Weight	< 500 g
Degree of protection	IP65
Operating temperature	–10 to +50 °C
Storage temperature	–20 to +70 °C
Ambient light limit	< 80 000 lux
Point cloud software	OleView

■ OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. The 50 m figure is the 80% reflectivity and reflector case; plan on 15 m at 10% reflectivity and 5 m at 4%. Repeat accuracy is specified from 0.4 m, not from the 0.1 m minimum range. Accuracy figures are standard values and depend on environmental conditions.

Installation. A 360° sensor needs its full circle clear — mount it above the platform's own structure rather than recessed into it. Both leads are 50 cm pigtails, so plan a junction close to the mount. Keep the optical window clear of spray and dust and clean it on a schedule in dirty environments.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-VFH150-2026.09 · Source: OLEI VFH1-50 2D LiDAR Sensor datasheet, Doc.# DSEN-VFH1-50-202510.